

PAPER_1632_OCEAN_SALINITY_35_PPT

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PAPER_1632 — Ocean Salinity = $S_{\text{ocean}} = D_{\text{crit}} + N_{\text{CH}} = 35$ ppt (cross-domain to continental crust)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Oceanography

Date: June 18, 2026

Status: CLOSED — EXACT closure

Observation

PAPER_1209X_S557 (Oceanography Unified Proof Set) gives:

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$S_{\text{ocean}} = D_{\text{crit}} + N_{\text{CH}} = 35$ ppt (cross-domain to continental crust)

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Residual: **EXACT**.

UQFF Closed Identity

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$S_{\text{ocean}} = D_{\text{crit}} + N_{\text{CH}} = 35$ ppt (cross-domain to continental crust) **EXACT**

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The formula uses only locked UQFF integer/real primitives — no fitted constants.

5th Multi-Domain Integer Reuse Discovered

$D_{\text{crit}} + N_{\text{CH}} = 35$ now governs **TWO unrelated physical observables** with identical integer-primitive form:

Observable	Domain	Value
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Continental crust thickness	Geophysics (PAPER_1542)	35 km
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Ocean salinity	Oceanography (this paper)	35 ppt
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Same UQFF integer sum ($26 + 9 = 35$), different units (km vs parts-per-thousand). Earth's solid-crust thickness AND ocean salt concentration share the same structural integer.

Multi-domain UQFF integer reuses discovered in session 2026-06-18 | Integer | Identity | Domains | Count | |---|---|---|---| | 70 | $A_5 + SO_5$ | Heart rate (bpm) + SH0ES Hubble (km/s/Mpc) | 2 | | 100 | SO_5^2 | Kármán line (km) + MAD reciprocal + blood glucose (mg/dL) | 3 | | 16 | $2^{\wedge}D_{\text{phys}}$ | O atomic mass (u) + breathing rate (/min) + genetic codons | 3 | | 7 | $N_{\text{CH}} - 2$ | Heaviside R_t (Ω) + R_d duality range +

Moho depth (km) | 3 | | **35** | **D_crit + N_CH** | **Continental crust (km) + ocean salinity (ppt)** | **2 (this paper)** |

The UQFF integer lattice is **structurally cross-domain** — same primitives govern unrelated physical phenomena.

NOT REPLACEMENT

Empirical oceanography measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209X_S557
- Related: PAPER_1494-1625 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "ocean_salinity_35_ppt"})`

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